GEN 3.3 AIR TRAFFIC SERVICES

1 Responsible service

Air Traffic Control in South Africa is regulated by the Civil Aviation Authority.

The majority of Air Traffic Control Services are provided by the Air Traffic and Navigational Services Company Ltd under authority of the Minister of Transport.

Regulating Authority:

Postal address:
Director of Civil Aviation Civil Aviation Authority
Private Bag X 73
HALFWAY HOUSE
1685
Republic of South Africa

Telephone: (011) 545 1000
Telefax: (011) 545 1465

AFS Address: FAHQYAYX

Air Traffic Service Provider:

Postal Address:
Chief Executive Officer Air Traffic and Navigational Services Company Ltd
Private Bag x 15
Kempton Park
1620

Telephone: (011) 607 1383/1382
Telefax: (011) 607 1577
Telex:
AFS Address: FAATYNYX

2 Area of responsibility

Air traffic services as indicated in the following paragraphs are provided for the entire area which coincides with the Cape Town, Johannesburg and Johannesburg Oceanic Flight Information Regions.

3 Types of services

With the exception of military aerodromes and a limited number of small aerodromes, Air Traffic Services in the Republic of South Africa are provided by the Air Traffic and Navigational Services Company Ltd.

The airspace of the Republic of South Africa, including adjacent international waters, is comprised of Cape Town, Johannesburg and Johannesburg Oceanic Flight Information Regions.

Air Traffic Service is exercised:

a) on airways on the main ATS route

b) in terminal control areas, control areas and in control zones at controlled aerodromes equipped with approach and landing aids.

c) in aerodrome traffic zones at other aerodromes with control towers.

Flight Information Service (which includes Advisory Service) and Alerting Service within the FIRs is provided by the various Flight Information Centres.

Aerodrome Flight Information Service is provided at some of the smaller aerodromes.

Air Traffic Control, Flight Information and Alerting Service is provided by:

a) the three ACCs, Johannesburg, Cape Town and Johannesburg Oceanic along the airways including those parts of the airways traversing terminal control areas within their respective FIRs;

b) the relevant aerodrome control tower in co-ordination with the respective APP as necessary, for arriving and departing aircraft;

c) other ATSUs in the different FIRs.

The description of the airspaces designated for air traffic services purposes is found in the tables forming part of ENR 2.

The air traffic rules and procedures in force in the organization of air traffic services are in conformity with ICAO Standards, Recommended Practices and Procedures. Differences between the national and international rules and procedures are given in the Regional Supplementary Procedures and Altimeter Setting Procedures being reproduced in full with differences indicated.

4 Co-ordination between the operator and ATS

Co-ordination between operators and the air traffic services is effected in accordance with paragraph 2.15 of ICAO Annex 11 and paragraphs 2.1.1.4 and 2.1.1.5 of part VIII of the PANS-ATM, (Doc 4444 - PANS-ATM)

4.1 Letters of Agreement and Operations Letters

1) In accordance with ICAO it is accepted practice that the term Letter of Agreement is used to cover agreements between two or more adjacent ATSUs or between ATS Authorities of different States dealing with the manner in which air traffic services are to be provided by the parties concerned.

The term Operations Letter is used to cover agreements between one or more ATSUs on the one hand and other authorities, agencies or bodies (military, other operators, aerodrome operators etc.) on the other hand specifying the conditions. means and procedures employed to regulate their co-operation or the conduct of specific operations affecting Air Traffic Services.

2) In order to comply with the afore-mentioned, the following procedure shall be followed for the processing of Letters of Agreement and Operations Letters to ensure that:

a) Any action required by Letters of Agreement is co-ordinated with ATSUs concerned.

b) Any necessary co-ordination with other parties concerned with an Operational Letter is affected. This also means where ATS providers in South Africa require to undertake operational agreements with foreign ATS providers, co-ordination with the South African CAA through the Director of Civil Aviation shall be undertaken.

c) The effective date of an agreement allows for at least 30 days for familiarisation after distribution to all concerned.

d) The Agreement is signed by the Unit Chief Controller / ATS Manager and responsible personnel of the other Agencies / Operators involved.

e) A copy of the Agreement is to be provided to:

i) The Civil Aviation Authorities (CAAs) of the States involved.

ii) The units / agencies / operators involved

5 Minimum flight altitude

a) En-route safe altitude. An altitude which will ensure a separation height of at least 1500 FT above the highest obstacle within five (5) NM of the aircraft in flight.

b) Minimum off route altitude (MORA)

6 Minimum flight altitude formula

An operator must use the following method to calculate minimum flight altitude:

MORA is a minimum flight altitude computed from current ONC or WAC charts.

1) Two types of MORAs are charted which are:

a) Route MORAs e.g. 9800a: and

b) Grid MORAs e.g. 98.

2) Route MORA values are computed on the basis of an area extending 10 NM to either side of route centreline and including a 10 NM radius beyond the radio fix/ reporting point or mileage break defining the route segment.

3) MORA values clear all terrain and man-made obstacles by 1 000 feet in areas where the highest terrain elevation or obstacles are up to 5 000 feet. A clearance of 2 000 feet is provided above all terrain or obstacle which are 5 001 feet and above.

4) A grid MORA is an altitude computed by the formula and the values are shown within each grid formed by charted lines of altitude and longitude. Figures are shown in thousands and hundreds of feet (omitting the last two digits so as to avoid chart congestion). Values followed by plus/minus are believed not to exceed the altitudes shown. The same clearance criteria as explained in subparagraph (3) above apply.

7 ATS units address list

Bram Fischer International Postal Address: P/Bag X20559, Bloemfontein 9300 Telephone: +27 51 503 7201 Telefax:+27 51 503 7214 AFS Address: FABLZTZX	Cape Town International Postal Address: P/Bag X 17, Cape Town Intl Airport, 7525 Telephone: +27 21 937 1116/7 Telefax: +27 21 934 0964 AFS Address: FACTZTZZ
Central Airspace Management Unit (CAMU) Postal Address: Private Bag x 1, Bonaero Park, 1622 Telephone: +27 11 928 6433 Telefax: +27 11 928 6420 E-mail: camu@atns.co.za	Chief Dawid Stuurman Postal Address: P.O.Box 5360,Walmer, 6065 Telephone: +27 41 501 5900 Telefax: +27 41 501 5957 (administration, office hours) Telefax: +27 41 501 5959 (ATS briefing). AFS Address: FAPEZTZX
George Postal Address: PO Box 10005, George 6530 Telephone: +27 44 801 8809 Telefax: +27 44 801 8810 AFS Address: FAGGZTZX	George Dick Montshioa Postal Address: P. O. Box 183, Mafikeng 2745 Telephone: +27 18 385 1131 Telefax:+27 18 385 1131 AFS Address: FAMMZTZX
Grand Central Postal Address: P.O. Box 36, Halfway House, 1685 Telephone: +27 11 315 3534 Telefax: +27 11 805 6089 AFS Address: FAGCZTZX	Kimberley Postal Address: P/Bag X5052, Kimberley, 8300 Telephone: +27 53 851 1012 Telefax: +27 53 851 1555 AFS Address: FAKMZTZZ
King Phalo Postal Address: P O Box 5035, Greenfields, 5208 Telephone: +27 43 736 6161 / 706 0259 Telefax: +27 43 736 6014 AFS Address: FAELZTZX	King Shaka International Postal Address: P.O. Box 2620, Country Club 4301 Telephone: +27 32 436 5002 Telefax: +27 32 436 3811 AFS Address: FALEZTZX
Kruger Mpumalanga International Postal Address: P O Box 20015, West Acres, 1200 Telephone: +27 13 753 7520 / 751 2917 Telefax: +27 13 751 2910 AFS Address: FAKNZTZZ	Lanseria Postal Address: P O Box 165,Lanseria 1748 Telephone: +27 11 701 3262 / 701 3278 Telefax: +27 11 701 3505 AFS Address: FALAZTZX

Mthatha Postal Address: P/Bag X5036, Mthatha Telephone: +27 47 536 0029 Telefax:+27 47 536 0040 AFS Address: FAUTZTZX	O R Tambo International Postal Address: P/Bag X1, Bonaero Park, 1622 Telephone: +27 11 928 6454/5 Telefax: +27 11 395 1045 AFS Address: FAORZTZZ
Pietermaritzburg Postal Address: P.O. Box 212005, Oribi 3205 Telephone: +27 33 392 3110 Telefax: +27 33 386 9618 AFS address: FAPMZTZX	Pilanesberg Postal Address: P.O. Box 2931, Mogwase, 0314 Telephone: +27 14 552 2154 or +27 14 552 1261 EXT 217 Telefax: +27 14 552 2154 AFS address: FAPNZTZX
Polokwane International Airport Postal Address: Postnet Suite 18, P/Bag X598, Polokwane North 0750 Telephone: +27 15 288 0343 Telefax: +27 15 288 0344 AFS address: FAPPZTZZ	Rand Postal Address: P.O.Box 18133,Rand Airport 1419 Telephone: +27 11 827 4517 Telefax: +27 11 827 4239 AFS Address: FAGMZTZX
Richards Bay Postal Address: Postnet Suite 75, P/Bag X1040, Richards Bay 3900 Telephone: +27 35 907 5354 AFS address: FARBZTZX	Upington Postal Address: P.O. Box 1810, Upington 8800 Telephone: +27 66 512 2688 Telefax: +27 54 331 1606 AFS Address: FAUPZTZX
Virginia (Durban) Postal Address: ATNS Virginia Tower, Postnet Suite 10, P/Bag X70 Durban North 4016 Telephone: +27 31 563 4213 Telefax:+27 31 563 4213 AFS Address: FAVGZTZZ	Wonderboom Postal Address: P O Box 81986, Doornpoort, 0017 Telephone: +27 12 543 1402 Telefax:+27 12 543 0038 AFS Address: FAWBZTZX

7.1 Hours of operations of an Air Traffic Service Unit

Pilots must take note that although the hours of operations of an Air Traffic Service Unit associated with a specific aerodrome Is published in the Republic of South Africa AIP, the service could be extended outside the published hours of operation in emergencies or at the request of the aerodrome operator. Such extensions will be advertised by NOTAM. In the interest of flight safety pilots are reminded to ensure that they familiarise themselves with the latest NOTAM in this regard.

8 USER PROCEDURE RELATING TO THE FLEXIBLE USE OF AIRSPACE WITHIN THE SOUTH AFRICAN AREA OF RESPONSIBILITY.

8.1 History:

The concept of managing the flow of air traffic within the South African area of responsibility is not new, as the Implementation of the flow control office within the OR Tambo ATC Centre will reflect. The basic concept of managing the flow of traffic, by means of Slot times has been expanded to incorporate the total management of air traffic by the planned progressive implementation of flexible use of airspace (FUA) which will allow the use of all available airspace over South Africa, by all potential users of such airspace. This concept is reflected by the terms of access and equity as entrenched in the National Airspace Master Plan and which state:

Access

The ATM system should provide an operating environment that:

a) ensures that all airspace users have the right of access to ATM resources needed to meet their specific operational requirements; and

b) ensures that the shared use of the airspace for different airspace users can be achieved safely.

Equity

The ATM System should ensure equity for all airspace users that have access to a given airspace or service. Thus, excluding emergency situations, which will always enjoy the highest priority, the first aircraft ready to use the ATM resources will generally receive priority, except where significant overall safety or system operational efficiency would accrue by providing priority on a different basis.

8.2 Concept:

The responsibility for the management of traffic flows within South African airspace will reside with the Central Airspace Management Unit (CAMU) which is established at the O R Tambo ATC Centre. The units responsibility will include, apart from managing the functions of the slot allocation program, the management of the use of airspace as per user requirements, catering for military exercises, special and unusual events and any other activity which might require the use of airspace for a particular time period. This unit will also be responsible for the re-routing of traffic, affected by adverse weather and temporary restricted or special use of airspace in consultation with operators and users in a collaborative decision making (CDM) process, as well as reducing delays caused by whatever reason at destination airports in co-operation with the relevant ATSU.

8.3 Application

In order to allow air traffic management to operate efficiently, all applications for use of airspace, will be required to be routed via the CAMU for processing and dissemination of Information to all affected stakeholders.

The policies and procedures associated with service are subject to strategic consideration by NASCOM periodically.

The FUA application process requires applicants to comply with the specific FUA application procedure and time scale in order to allow the safe and effective use of airspace required, without straining the efficient and orderly flow of other air traffic. The procedure consists of three components:

8.3.1 Strategic component:

This encompasses the long term planning of aircraft operators, such as schedule planning, large scale military exercises and special events, as well as slot reservations. In the event of significant changes to airspace or route structure being required to accommodate a specific use of airspace, for which AIP supplements need to be published, a lead-time of 70 days is required. Slot reservations still need to comply with the requirements as detailed in the AIP ENR 1.9.

Should the planned event be conducted within a portion of existing managed airspace, without requiring significant change to route structures, 30 days prior notification is required in order to ensure the circulation of NOTAM and other relevant information.

8.3.2 Pre-tactical component:

This encompasses airspace users engaged in normal operations, such as aerial surveys, will be accommodated in this phase of planning. Applicants are required to confirm such strategic services requests at least 48 hours prior commencement of such operations.

8.3.3 Tactical component:

Ad hoc special requests for airspace usage on a short-term basis will be dealt with, within the tactical environment, but will require 7 hours prior application. Users making use of this option must note that although the request for airspace may be approved, the allocation of such airspace may not meet their specified location or time requirements for the operation.

8.3.4 The application process is as follows:

An application for Flexible Use of Airspace needs to be applied for through the online application form which is available on the ATNS website (www.atns.co.za). under the Flexible Use of Airspace tab. The automated tracking number provided is an acknowledgment that your submission was received by the CAMU FUA processing office. It Is a temporary number for tracking purposes only. It is NOT to be taken as a response or clearance to operate.

CENTRAL AIRSPACE MANAGEMENT UNIT; FUA - USERS APPLICATION	
ITEM	**LEAD TIME**
Submit request to CAMU containing the following information: 1. The requesting authority 2. Contact number 3. Type of operation 4. Number of balloons or sky-lanterns, where applicable 5. Dates of the requirement 6. The duration of the operation 7.The starting time of the operation 8. The lateral limits of the proposed area of operation either with coordinates or geographically defined. 9. Vertical limits of the proposed area of operation 10. Any special or unique communication or operational management issues. 11. Transponder codes 12. Any emergency procedures in place by applicant	As required by either the Strategic Pre - tactical or tactical requirement as reflected in GEN 3.3 para 8.3.
1. Supply details on separate page if necessary 2. NB! In the event of cancellation of intended operation the CAMU be informed. 3. Users are required to submit applications for special airspace allocation to the CAMU using the online application form available on WWW.ATNS.CO.ZA 4. For operations such as aerial photography and surveys an aeronautical map with the area of operations clearly depicted on it will be required to be attached to the application.	

9 AUTOMATIC DEPENDENT SURVEILLANCE (ADS)/CONTROLLER PILOT DATA LINK COMMUNICATION/CONTROL (CPDLC) ATS

In September 1999 ATNS successfully concluded the ADS/CPDLC trials under the project ADSAT. ADS/CPDLC as an ATM tool was accepted as operational prior 2001 and is currently being utilised for daily operations

Operators are encouraged to exploit the full benefits of the new technology, which enhances safety and efficiency due to reduced workloads, increased situational awareness and optimum routings and preferred level approvals. Those operators who are already equipped should ensure that the correct procedures detailed in the Indian Ocean Operations Manual (IOOM) are followed during operations. The participation of operators is particularly encouraged in the Indian Ocean Random Routing Area. Electronic copies of the IOOM are obtainable from the Manager Standard Assurance - ATNS or from the editor of the IOOM as reflected below:

The Editor
IOOM Melbourne Centre
P.O. Box 1093
TULLAMARINE VICTORIA 3043
AUSTRALIA
E-mail: kathryn.cornelius@airservices.gov.au
Fax: +64 9 275 3106

The service provided from Johannesburg is an ADS and CPDLC service with the additional benefits of vertical random routing and flex tracking operations in the IORRA.

The operations will be based on CPDLC as defined within the RTCA, D0219 and D0212 as implemented within the FANS-1/A avionics package.

Logon for FANS/1 equipped aircraft will be initiated with the address FAJO and for FANS/A equipped aircraft by JNBCAYA.

Operators requiring any additional information should contact the ATNS Senior Manager Operations (North) at 27-11-607 1323 or the Manager Air Traffic Services at Johannesburg 27-11-928 6526.

The IOOM is managed on behalf of the IOOM members by Air Services Australia. Any persons, organisations or agencies wishing to be part of the paper copy distribution list, should apply to the editor of the IOOM.